LHCb Open Data Analysis of Three-Body B-Meson Decays

Invariant-mass reconstruction, signal classification, CP-asymmetry diagnostics and Dalitz-plot studies

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HEP portfolio project

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Physics Motivation

- The visible Universe is matter dominated.
- CP violation is required but not yet sufficient in the Standard Model.
- Heavy-flavour decays probe weak phases and hadronic interference.

Why B mesons?

- many decay channels
- displaced vertices
- rich multibody phase space

LHCb Experiment and Dataset

- LHCb: forward spectrometer at the LHC.
- Optimized for beauty and charm hadrons.
- Excellent tracking, vertexing and particle identification.

Input sample

- CERN Open Data
- ROOT files
- `DecayTree` TTree
- track momenta, charge, PID, vertex variables

Analysis Workflow

ROOT → `uproot` → `pandas` → reconstruction → cuts → ML → physics observables

Data

Branch loading
candidate tables
reproducible paths

Physics

invariant mass
event selection
charge split

Outputs

figures
ML diagnostics
CP and Dalitz studies

Invariant-Mass Reconstruction

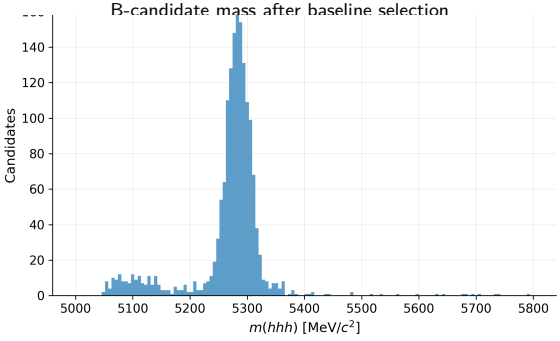
Three-kaon hypothesis

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_K^2}$$

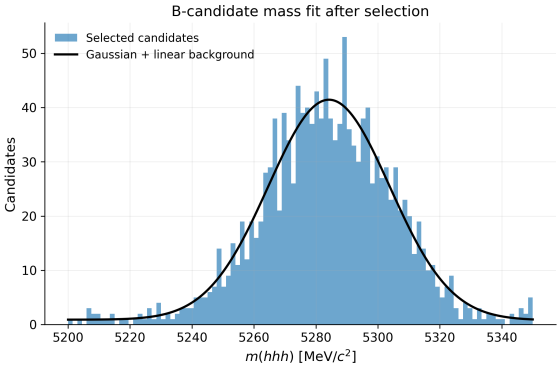
$$m_B^2 = \left(\sum_i E_i \right)^2 - \left| \sum_i \vec{p}_i \right|^2$$

- assign $m_K = 493.677\ \mathrm{MeV}/c^2$
- build daughter four-vectors
- sum to reconstruct B candidate
- vectorized in `numpy`/`pandas`

Mass Spectrum and Signal Fit



Selected candidates



Gaussian + linear background

$$f(m) = Ae^{-(m-\mu)^2/(2\sigma^2)} + am + b$$

Event Selection

Baseline cuts

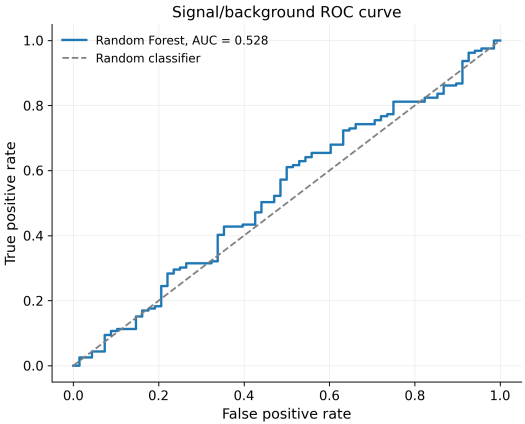
$$\chi^2_{\text{vtx}}(B) < 5$$

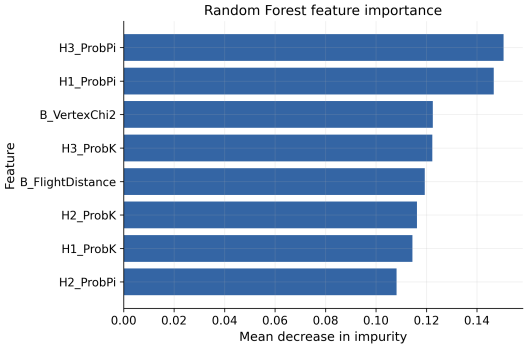
$$d_{\text{flight}}(B) > 20$$

$$P_K(H_1), P_K(H_2), P_K(H_3) > 0.5$$

- vertex quality: common displaced decay point
- flight distance: B lifetime signature
- kaon PID: consistency with $B \to KKK$
- same cuts reused across notebooks

Signal/Background Classification





- Sideband-labelled Random Forest classifier.
- AUC is a diagnostic of separation power, not the main physics result.

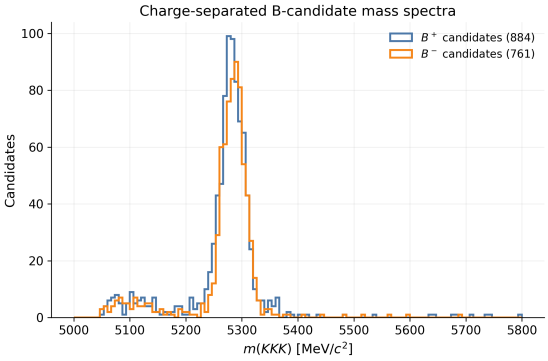
The low AUC indicates limited residual separation after the baseline cuts, not a failure of the physics reconstruction.

CP-Asymmetry Diagnostic

$$A_{\mathrm{CP}}^{\mathrm{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}$$

- signal window: $5270 < m_B < 5290\ \mathrm{MeV}/c^2$
- $N(B^+) = 282, \quad N(B^-) = 246$
- $A_{\mathrm{CP}}^{\mathrm{raw}} = -0.06818$

Not corrected for detector or production effects.



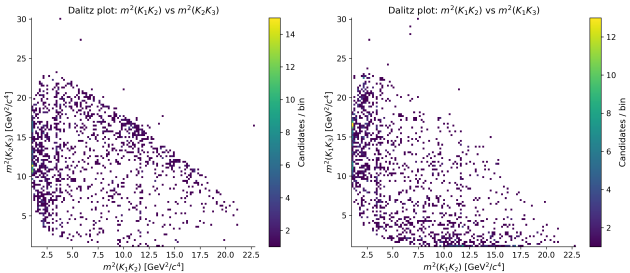
Dalitz-Plot Analysis

- Pairwise invariant masses:

$$m_{ij}^2 = (E_i + E_j)^2 - |\vec{p}_i + \vec{p}_j|^2$$

- $m^2(K_1K_2)$
- $m^2(K_2K_3)$
- $m^2(K_1K_3)$

Useful for resonance and local CP studies.



Limitations and Future Work

Current limitations

- raw asymmetry only
- single-polarity demonstration
- simple mass model
- sideband proxy labels

Next steps

- combine MagnetUp/MagnetDown
- detector corrections and simulation
- XGBoost or boosted decision trees
- local CP asymmetries in Dalitz bins

Conclusion

- Reproducible HEP workflow from ROOT ntuples to physics plots.
- Invariant-mass reconstruction under a kaon mass hypothesis.
- Shared selections and modular Python functions.
- Sideband ML diagnostics, raw CP-asymmetry study and Dalitz plots.

A compact portfolio analysis mirroring a real LHCb workflow